Homework — 1.4.2 Data Structures — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Total: Part A 14 + Part B 6 = 20 marks.

Part A

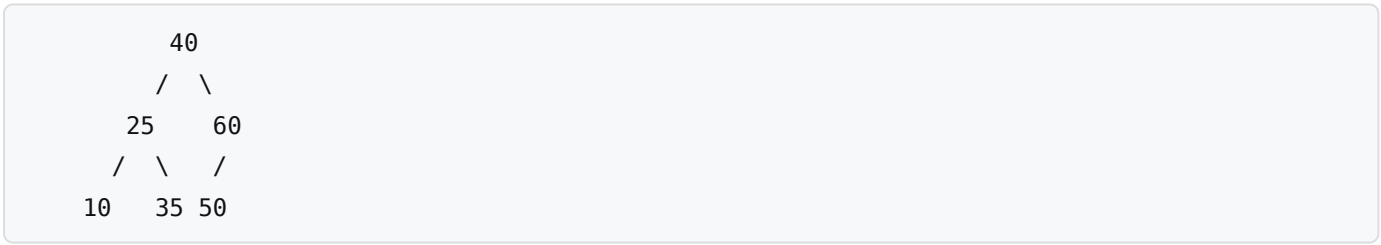
1. [4] Stack trace (top listed first): - push(8) → [8] - push(3) → [3, 8] - push(6) → [6, 3, 8] - pop() → returns **6** → [3, 8] - pop() → returns **3** → [8] - push(1) → [1, 8]

Values popped (in order): **6, then 3**. Final stack (top → bottom): **1, 8**. *Mark: pushes added to the top in correct order (1); first pop returns 6 (1); second pop returns 3 (1); correct final contents 1, 8 with 1 on top (1).*

2. [3] One mark per point: - A **new node** is created (in free memory) containing the value 15 (1). - The new node's pointer is set to point to the node containing **20** — i.e. it is given the value currently held in node 12's pointer (1). - The pointer of the node containing **12** is updated to point to the new node (1).

The existing nodes are not moved — only pointers change. Accept the steps in any workable order, but the new node must be linked to 20 using node 12's original pointer before (or as) that pointer is overwritten. (AO2)

3. [3] BST after inserting 40, 60, 25, 35, 10, 50:



- 40 is the root.
- 60 > 40 → right of 40; 25 < 40 → left of 40.
- 35 > 25 (and < 40) → right of 25; 10 < 25 → left of 25.
- 50 < 60 (and > 40) → left of 60. *Mark: 40 root with 25 left / 60 right (1); 10 left and 35 right of 25 (1); 50 as left child of 60 (1).*

4. [2] In-order traversal = **Left, Root, Right**: **10, 25, 35, 40, 50, 60** (1). Special property: an in-order traversal of a **BST** visits the nodes in **ascending (sorted) order** (1). *(AO2)*

5. [2] 23 MOD 10 = **3**; 57 MOD 10 = **7**; 33 MOD 10 = **3** (1). **Collision:** 33 hashes to index 3, which is already occupied by 23 (1 — for identifying the collision and naming a resolution). Resolution method (any one): **linear probing** (place at the next free slot) **or chaining** (store a linked list of keys at that index). *Mark: all three indices correct 3, 7, 3 (1); collision at index 3 identified AND a valid resolution method named (1).*

Part B

6. [2] Interrupts can occur while another interrupt is already being serviced (nested), so the CPU must be able to return to each task in the correct order (1). A stack is **LIFO**, so the registers (e.g. PC and ACC) of the interrupted task are **pushed** on, and when the higher-priority ISR finishes they are **popped** off in reverse order — the most recently interrupted task resumes first — guaranteeing every task resumes correctly even with several interrupts nested (1). *(AO1)*

7. [1] The control bus carries **control signals** (e.g. read/write signals, clock pulses, interrupt requests) between the CPU and other components to coordinate/synchronise their operation. *(AO1)*

8. [3] A foreign key is an attribute in one table that is the **primary key of another table**, used to **link** the two tables together (1). Referential integrity ensures that every foreign-key value **matches a valid existing primary key** in the related table (or is null) (1), so there are **no "orphaned" records** that reference a row which does not exist (1). *(AO1)*